

Telco Customer Churn Dataset - Complete Documentation

Dataset Overview

This dataset contains customer information from a telecommunications company, focusing on customer retention and churn analysis. It includes 21 features describing customer demographics, services, account information, and the target variable indicating whether a customer has churned.

Business Context

This is a customer churn prediction dataset for a telecommunications provider offering:

- Phone services (landline)
- Internet services (DSL, Fiber optic)
- Various add-on services (security, backup, streaming, etc.)
- Different contract terms and payment methods

The primary business objective is to identify customers at risk of churning to implement retention strategies.

Column Documentation

Customer Identification

- **customerID:** Unique identifier for each customer
 - **Data Type:** String
 - **Format:** 4 digits followed by hyphen and 5 letters (e.g., "7590-VHVEG")
 - **Business Use:** Customer tracking and record linking

Demographics

- **gender:** Customer's gender
 - **Values:** Male, Female
 - **Business Use:** Demographic segmentation and targeted marketing
- **SeniorCitizen:** Whether customer is a senior citizen
 - **Values:** 0 (No), 1 (Yes)
 - **Note:** Numeric encoding instead of Yes/No like other binary fields
 - **Business Use:** Age-based service offerings and pricing strategies
- **Partner:** Whether customer has a partner/spouse
 - **Values:** Yes, No
 - **Business Use:** Family plan opportunities and stability indicators

- **Dependents: Whether customer has dependents**
 - **Values: Yes, No**
 - **Business Use: Family service bundles and churn risk assessment**

Account Information

- **tenure: Number of months customer has been with company**
 - **Data Type: Integer (0-72 months)**
 - **Range: 0-72 months observed in dataset**
 - **Business Use: Customer lifetime value, loyalty segmentation**

Service Offerings

Core Services

- **PhoneService: Whether customer has phone service**
 - **Values: Yes, No**
 - **Business Use: Service bundle analysis**
- **MultipleLines: Multiple phone lines status**
 - **Values: Yes, No, "No phone service"**
 - **Dependency: Only applicable when PhoneService = Yes**
 - **Business Use: Upselling opportunities**
- **InternetService: Type of internet service**
 - **Values: DSL, "Fiber optic", No**
 - **Business Use: Service tier analysis, infrastructure planning**

Add-on Services (Internet-dependent)

The following services are only available with internet service:

- **OnlineSecurity: Online security service**
 - **Values: Yes, No, "No internet service"**
 - **Business Use: Security service adoption, revenue per customer**
- **OnlineBackup: Online backup service**
 - **Values: Yes, No, "No internet service"**
 - **Business Use: Data service revenue tracking**
- **DeviceProtection: Device protection service**

- **Values:** Yes, No, "No internet service"
 - **Business Use:** Insurance revenue, customer care costs
- **TechSupport:** Technical support service
 - **Values:** Yes, No, "No internet service"
 - **Business Use:** Support cost analysis, service quality
- **StreamingTV:** TV streaming service
 - **Values:** Yes, No, "No internet service"
 - **Business Use:** Entertainment service adoption
- **StreamingMovies:** Movie streaming service
 - **Values:** Yes, No, "No internet service"
 - **Business Use:** Content service revenue

Contract & Billing

- **Contract:** Contract term length
 - **Values:** "Month-to-month", "One year", "Two year"
 - **Business Use:** Customer commitment levels, churn risk assessment
- **PaperlessBilling:** Whether customer uses paperless billing
 - **Values:** Yes, No
 - **Business Use:** Cost reduction initiatives, customer preferences
- **PaymentMethod:** How customer pays bills
 - **Values:** "Electronic check", "Mailed check", "Bank transfer (automatic)", "Credit card (automatic)"
 - **Business Use:** Payment processing costs, automation levels

Financial Information

- **MonthlyCharges:** Monthly bill amount
 - **Data Type:** Numeric (currency)
 - **Range:** Varies based on services
 - **Business Use:** Revenue analysis, pricing optimization
- **TotalCharges:** Total amount charged over tenure
 - **Data Type:** Numeric (currency)

- **Note:** Contains some blank values for new customers (tenure = 0)
- **Calculation:** Should approximate $\text{MonthlyCharges} \times \text{tenure}$
- **Business Use:** Customer lifetime value, revenue forecasting

Target Variable

- **Churn:** Whether customer has left the company
 - **Values:** Yes (churned), No (active)
 - **Business Use:** Primary prediction target for retention models

Key Performance Indicators (KPIs)

Customer Metrics

- 1. Churn Rate:** Percentage of customers who cancelled service
 - **Formula:** $(\text{Churned Customers} / \text{Total Customers}) \times 100$
 - **Industry Benchmark:** 15-25% annually for telecom
- 2. Customer Lifetime Value (CLV):** Expected revenue from customer
 - **Formula:** $(\text{Average Monthly Revenue} \times \text{Gross Margin \%}) / \text{Monthly Churn Rate}$
 - **Use:** Investment decisions in customer acquisition/retention
- 3. Average Revenue Per User (ARPU):** Monthly revenue per customer
 - **Formula:** $\text{Total Monthly Revenue} / \text{Total Active Customers}$
 - **Use:** Pricing strategy and market positioning

Retention Metrics

- 4. Customer Tenure Distribution:** How long customers stay
 - **Segments:** 0-12 months (new), 13-36 months (growing), 37+ months (mature)
 - **Use:** Identify critical retention periods
- 5. Contract Mix:** Distribution of contract types
 - **Use:** Revenue predictability and churn risk assessment
- 6. Service Penetration Rates:** Adoption of each add-on service
 - **Formula:** $(\text{Customers with Service} / \text{Total Customers}) \times 100$
 - **Use:** Upselling opportunities and bundle effectiveness

Financial KPIs

- 7. Monthly Recurring Revenue (MRR):** Predictable monthly income

- **Use: Business health and growth tracking**

8. Revenue per Service: Income contribution by service type

- **Use: Service profitability analysis**

9. Payment Method Distribution: Customer payment preferences

- **Use: Processing cost optimization**

Operational KPIs

10. Service Bundle Penetration: Multiple services per customer

- **Use: Cross-selling effectiveness**

11. Digital Adoption Rate: Paperless billing percentage

- **Use: Operational cost reduction tracking**

12. Senior Citizen Retention: Churn rates for senior customers

- **Use: Specialized retention programs**

Business Insights Framework

Churn Risk Factors

- **High Risk: Month-to-month contracts, electronic check payments, newer customers (< 12 months)**
- **Medium Risk: Single service customers, no add-ons, high monthly charges relative to services**
- **Low Risk: Long-term contracts, automatic payments, multiple services, longer tenure**

Revenue Optimization

- **Upselling Targets: Internet customers without add-on services**
- **Contract Conversion: Month-to-month to annual contracts**
- **Payment Automation: Manual payment customers to automatic methods**

Customer Segmentation

- 1. High-Value Loyalists: Long tenure + high ARPU + low churn risk**
- 2. Growth Prospects: Medium tenure + room for service expansion**
- 3. At-Risk Churners: Short tenure + minimal services + month-to-month**
- 4. Price Sensitive: High churn correlation with pricing changes**

Data Quality Notes

- **Missing Values:** TotalCharges has blank entries for some new customers
- **Encoding Inconsistency:** SeniorCitizen uses 0/1 while other binaries use Yes/No
- **Logical Dependencies:** Add-on services depend on internet service availability
- **Calculation Verification:** TotalCharges should align with MonthlyCharges × tenure

Recommended Analysis Approaches

1. **Cohort Analysis:** Track customer behavior over time segments
2. **Survival Analysis:** Time-to-churn modeling
3. **RFM Analysis:** Recency, Frequency, Monetary customer segmentation
4. **Price Sensitivity Analysis:** Churn correlation with pricing tiers
5. **Service Bundle Analysis:** Optimal service combinations for retention